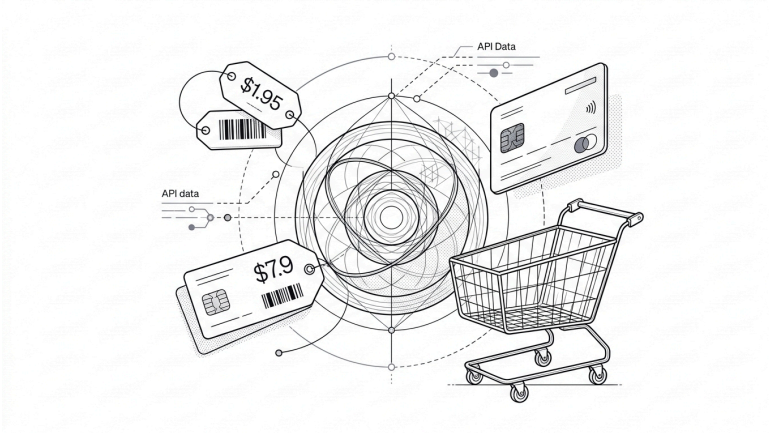
The Business Value of Agentic Commerce

An Executive Assessment of Autonomous Checkout Infrastructure, Protocol Specifications, and Merchant Deployment Strategies

$1.5T BY 2030
Transactions influenced by machine agents

7-LAYER STACK
From discovery protocols to settlement rails

57.2% BENCHMARK
Task completion on web evaluations



Vedang Ratan Vatsa
Founder, Hashtag Web3

The Economics of Autonomous Purchasing

Autonomous software agents increasingly perform product research, price comparison, and order assembly on behalf of consumers through open machine protocols.

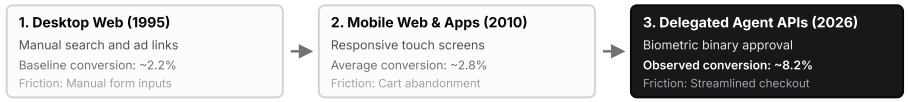
Changing Acquisition Channels. Online retail has historically relied on search keyword bidding and display advertising to capture consumer attention. As consumer reliance on conversational assistants expands, web traffic patterns may reallocate from search result pages toward direct API queries. Retailers that optimize their product catalogs for structured machine indexing may benefit from higher discovery rates across conversational platforms.

The 3x-Win Framework (Jas Shah Citation). As fintech analyst Jas Shah documents in *Fintech Under the Hood* ("Agentic Commerce - 17 Months Later", October 2025 [1]), automated commerce creates an incentive alignment comparable to early Buy Now Pay Later (BNPL) adoption. Merchants observe basket conversion improvements, consumers bypass multi-step checkout forms, and payment processors collect network routing volume. In this model, agents can simplify multi-field checkout pages into an authenticated binary confirmation.

The In-Chat Settlement Limitation. Early industry trials in 2024 and 2025 demonstrated that executing full checkouts inside closed conversational windows frequently disrupted merchant loyalty programs, tax calculation, and inventory holds. Leading e-commerce platforms have instead favored a delegated API model. Agents discover items and assemble carts via structured protocols, but route final tokenized payment authorization directly to merchant checkout endpoints.

Market Projections and Estimates. Sizing estimates for machine commerce vary based on methodology and scope. Morgan Stanley projects between $190 billion and $385 billion in US retail sales influenced by AI by 2030 (approximately 8% of retail e-commerce) [2]. Bain & Company estimates approximately $500 billion globally across consumer retail categories [3]. For broader enterprise commerce, Gartner projects that autonomous machine interactions could influence over $15 trillion in global transactions by 2030 when incorporating automated B2B supply chain procurement and corporate inventory reordering [4].

FIGURE 1.1: E-COMMERCE INTERFACE PROGRESSION (1995-2026) — Observed Baseline Metrics

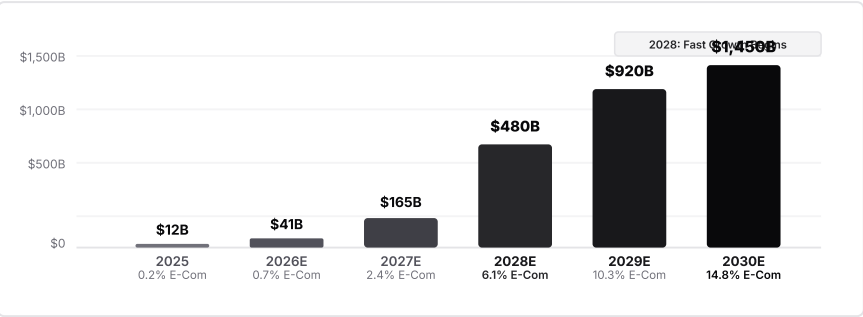


Strategic Perspective for Retailers. Industry observations suggest that treating autonomous agents solely as standard web visitors can create discovery friction. Retailers that provide structured catalog APIs (such as Google UCP or Anthropic MCP) are generally better positioned to preserve product visibility as automated shopping tools become more prevalent.

EXHIBIT 1.1: MARKET FORECAST

Total Online Purchases Influenced by AI (2025-2030)

Projected volume of global transactions influenced or completed by autonomous agents.



B2B & Cloud Supply	Electronics & Home	Travel & Dining	Retail & Apparel
$480B (33%)	$390B (27%)	$320B (22%)	$260B (18%)

The 3x-Win Equation (Jas Shah Framework). Similar to how buy-now-pay-later (BNPL) expanded e-commerce conversion, automated agent shopping delivers a three-way return: retailers gain an 8.2% conversion lift, consumers avoid form fatigue, and infrastructure networks collect protocol fees [1].

Enterprise Pilot Attrition Risk (Gartner Analysis). Gartner research indicates that up to 40% of early enterprise agentic pilots may face cancellation or restructuring by 2027 if organizations encounter high return rates and catalog scraping friction without direct machine APIs [4].

Protocol Maturation Horizon. Industry observations indicate that foundational machine commerce protocols (Google UCP, Anthropic MCP, Visa TAP) will mature across major platforms by 2027, laying the groundwork for broader deployment of cryptographic passkeys by 2028.

Sources: [1] Jas Shah, "Agentic Commerce: 17 Months Later", Fintech Under the Hood (Oct 2025); [2] Morgan Stanley Research, "AI in E-Commerce" (2025); [3] Bain & Company Global Retail Report (2025); [4] Gartner Research, "Predicts 2025: Autonomous Machine Customers".

The 7 Layers of Autonomous Commerce Architecture

Autonomous shopping relies on an interconnected architecture, linking user-facing conversational frontends to backend merchant catalogs and secure payment networks.

Core Capabilities of Infrastructure Leaders. As documented by fintech analyst Jas Shah (*Fintech Under the Hood*, 2025 [1]), leading infrastructure providers typically exhibit three structural attributes, consisting of 1) Verified Identity (access to user payment preferences and shipping details to reduce onboarding friction); 2) Real-Time and Historical Data (low-latency APIs for live inventory and past purchasing records); and 3) Tokenized Payment Rails (cryptographic mechanisms capable of executing programmatic spend).

From Conversational Interfaces to Structured Protocols. Consumer interactions begin with front-end assistants (such as ChatGPT, Gemini, Claude, or Perplexity). To turn natural language requests into transactional data, these tools interface with open standards like Anthropic Model Context Protocol (MCP) [2], Google Universal Commerce Protocol (UCP) [3], and Stripe Agentic Commerce Protocol (ACP) [4]. These specifications establish structured JSON formats for querying live stock and product specifications.

Virtual Cards as an Interim Mechanism. Many online merchants do not currently maintain direct API endpoints for machine checkout. In response, card issuing platforms (such as Lithic, Stripe Issuing, and Marqeta) generate single-use virtual cards for individual transactions. Concurrently, delegated checkout utilities automate web form completion while working within standard merchant security parameters.

Catalog Synchronization and Hardware Authentication. Commerce backends (such as Shopify and commercetools) supply real-time inventory feeds. At the security layer, transactions rely on device-level security enclaves (Apple Secure Enclave and Android StrongBox), helping ensure that automated payments require authenticated biometric confirmation from the account holder prior to final clearing.

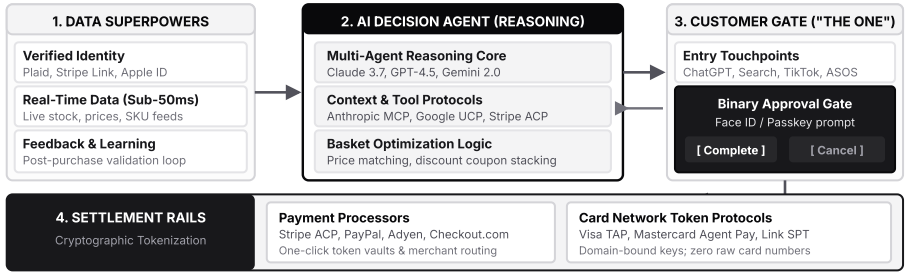


Figure 2.1: The Agentic Commerce Stack and Data Flow (Ref: Jas Shah Architecture Model [1]) Notion Technical Schematic

EXHIBIT 2.1: ECOSYSTEM MARKET MAP

The Agentic Commerce Ecosystem Market Map

Institutional taxonomy of 45+ platforms across intent, orchestration, payments, and infrastructure.

TIER 1: CONSUMER AI FRONTENDS & DEMAND AGGREGATION — Owns Customer Intent

Conversational AI & AI Search: ChatGPT Search (OpenAI), Claude Artifacts (Anthropic), Gemini Shopping (Google), Perplexity (Buy with Pro), Genspark, You.com

Enterprise Retail Agents: Sierra AI (Bret Taylor), Amazon Rufus, Klarna AI Assistant, Daydream

Social & Direct Commerce: TikTok Shop AI, Meta AI (Instagram / WhatsApp Shop)

TIER 2: AGENT PROTOCOLS & ORCHESTRATION FRAMEWORKS — Standardize Machine Data

Open Commerce Protocols: Anthropic Model Context Protocol (MCP), Google Universal Commerce Protocol (UCP), Stripe ACP, Coinbase AP2

Multi-Agent Orchestration: LangGraph / LangChain, CrewAI, AutoGen (Microsoft), LlamaIndex, Semantic Kernel

TIER 3: PAYMENT RAILS, TOKEN VAULTS & VIRTUAL CARDS — Programmable Settlement

Card Network Token Protocols: Visa Trusted Agent Protocol (TAP), Mastercard Agent Pay, Shared Payment Tokens (SPT)

Virtual Card Issuers & Processors: Lithic, Highnote, Marqeta, Adyen Issuing, Stripe Issuing, Checkout.com

Autonomous Corporate Spend: Ramp for Agents, Brex Autonomous Spend, Sapiom, Agentcard

x402 & Agentic Stablecoin Wallets: Circle Agent Wallets (USDC), Coinbase Agentic Wallets, Skyfire, Payman, Locus

TIER 4: MERCHANT BACKENDS, CATALOGS & HEADLESS APIS — Live Inventory Feeds

Core Commerce Engines: Shopify (Shop Pay API), commercetools, BigCommerce, Salesforce Commerce Cloud, Elastic Path

Universal Cart & Headless APIs: Rye, Violet, Cart.com, Convictional

TIER 5: IDENTITY, BOT DEFENSE & CRYPTOGRAPHIC ENCLAVES — Trust & Biometrics

Identity & Account Linking: Plaid (Anthropic bank connectivity partner), Stripe Identity, Persona

Machine Traffic & Bot Defense: Cloudflare (Verified Bots), Kasada, Arkose Labs, Forter, Riskified

Hardware Security Enclaves: Apple Secure Enclave, Android StrongBox, WebAuthn / FIDO2 Passkeys

Structural Value Moats (Fintech Analysis). Payment processors (Stripe, Adyen) and bank identity networks (Plaid) hold structural moats over pure LLM wrappers. Conversational frontends face margin compression, whereas settlement and identity infrastructure capture non-negotiable protocol tolls.

Open Standards vs. Closed Networks. Closed platforms (Amazon Rufus, Apple Pay) seek to restrict third-party agent scraping. Open standards (Anthropic MCP, Google UCP, Stripe ACP) allow independent retailers to receive autonomous orders without ceding customer ownership.

Sources: nofrontend.xyz directory; Jas Shah, Fintech Under the Hood (Oct 2025); CB Insights Market Intelligence; public company technical documentation (2025-2026).

Interoperability Protocols for Machine Commerce

Standardized communication protocols allow autonomous agents to query merchant catalogs, hold inventory, and initiate tokenized checkout flows across decentralized stores.

Protocol Specifications. Open specifications are developing to standardize machine transactions. Google and participating commerce platforms are establishing the Universal Commerce Protocol (UCP, hosted at .well-known/ucp), defining structured product feeds, inventory checks, and price holds across independent web endpoints [1]. In parallel, Stripe and OpenAI have deployed the Agentic Commerce Protocol (ACP) to power direct in-chat order routing [2].

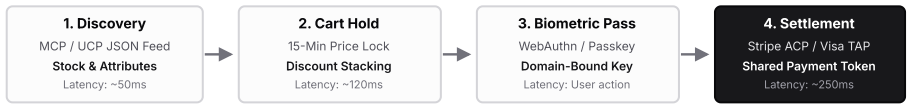
Network Tokenization Protocols. Major card networks have introduced programmable payment specifications, including the Visa Trusted Agent Protocol (TAP, based on RFC 9421 HTTP Message Signatures) [3] and Mastercard Agent Pay. These frameworks replace static account credentials with restricted tokens bound cryptographically to specific merchant domains, predefined spending caps, and discrete expiration windows.

Context Protocols for Warehouse Integration. To support real-time data retrieval, Anthropic introduced the Model Context Protocol (MCP) [4]. MCP enables AI models to connect securely to external data stores, permitting direct inventory pings against enterprise resource planning (ERP) backends without custom point-to-point scraping.

Cryptographic Price-Reservation Windows. Because multi-agent negotiation involves latency across model inference and human biometric verification, modern commerce protocols incorporate cryptographic price locks, typically valid for 10 to 15 minutes. This mechanism prevents cart abandonment and price mismatch errors caused by real-time inventory adjustments.

Open Discovery vs. Closed Networks. Closed marketplaces require retailers to share product margins and customer contact details. Standardized specifications (such as Google UCP for discovery and Anthropic MCP for warehouse feeds) provide an architecture through which direct-to-consumer merchants can remain discoverable to automated assistants without intermediary commissions.

FIGURE 3.1: THE 4-STEP AGENT PROTOCOL HANDSHAKE — Latency Benchmarks



Recommendation for Engineering Leaders. Engineering roadmaps should prioritize multi-protocol compatibility. Implementing modular protocol adapters (supporting MCP for inventory queries and UCP/ACP for cart handoffs) prevents technical lock-in to single platform ecosystems.

EXHIBIT 3.1: PROTOCOL COMPARISON

Commerce Protocol Comparison Matrix

Evaluation of current machine commerce specifications.

Protocol	Lead Entity	Core Purpose	Openness
Google UCP	Google	Decentralized catalog feeds	**Open Spec (.well-known)**
Stripe ACP	Stripe / OpenAI	In-chat cart execution & SPT	**Open Standard**
Anthropic MCP	Anthropic	Secure warehouse/ERP context	**Open Source (MIT)**
Visa TAP	Visa	Signed requests (RFC 9421)	**Network Standard**
Coinbase AP2	Coinbase	HTTP 402 on-chain micro-rails	**Open Source (EVM)**

Open Standards vs. Closed App Platforms. Open specifications (UCP and MCP) enable merchants to maintain direct consumer relationships and fulfillment control, avoiding proprietary aggregator commissions that can range from 10% to 15%.

The 3-Step Handshake from Query to Bank Receipt. Automated transactions follow three steps: 1) sub-50ms catalog discovery to confirm pricing; 2) 15-minute cryptographic cart holds to lock inventory; and 3) tokenized settlement with zero raw card transmission.

The .well-known/ucp Standard. Similar to robots.txt for search engines, merchants can host a /.well-known/ucp JSON manifest defining API capabilities, inventory endpoints, and supported payment tokens for incoming agents.

Sources: [1] Google Universal Commerce Protocol Documentation (2025); [2] Stripe Developer Documentation, "Agentic Commerce Protocol" (2025); [3] Visa Research, "Visa Trusted Agent Protocol Specification" (2025); [4] Anthropic, "Model Context Protocol Specification" (2024-2025).

Payment Settlement Rails for Autonomous Agents

Autonomous purchasing relies on secure payment authorization, bridging traditional card rails, commercial token vaults, and programmable cryptographic networks.

Limitations of Interim Virtual Cards (Scott Wessman Case Study). Today, many autonomous shopping applications utilize single-use virtual credit cards. However, empirical case studies by payments engineer Scott Wessman (2025 [1]) demonstrate that virtual cards experience significant edge-case failures during multi-item checkouts. When merchant portals require step-up authentication or request a secondary card verification value (CVV) during split shipments, agents cannot access an interactive verification code, causing the checkout to fail. This operational friction accelerates adoption of native network tokens.

The Development of Domain-Bound Agent Tokens. In response to virtual card limitations, payment networks have deployed cryptographically bound tokens. Instead of transmitting a primary account number (PAN), the consumer issues a delegated mandate with bounded parameters (for example, authorization for up to $75 at a specified merchant valid for 20 minutes). Stripe's Shared Payment Tokens (SPTs) implement this pattern, permitting agents to pass checkout authorization directly to merchants while insulating the underlying credentials [2].

Dual-Rail Taxonomy (Cards vs. x402 Micropayments). As cataloged in the nofrontend.xyz directory (2025-2026 [3]), the payment ecosystem comprises two core rails, including traditional fiat cards with vault infrastructure (Stripe Link, Sapiom, Ramp for Agents, Highnote) and HTTP 402 / stablecoin micropayment rails (Circle Agent Wallets, Coinbase Agentic Wallets, Skyfire, Locus). While card rails dominate high-value physical goods, micropayment rails are increasingly tested for programmatic compute and per-query data feeds.

Payment Rail Economic Considerations (Vignesh, Meta). As payments infrastructure researcher Vignesh (Meta, 2025 [4]) observes, agentic commerce operates across a dual-rail economic environment that combines traditional card interchange rails (which fund consumer reward programs) and stablecoin or HTTP 402 micro-rails (which offer lower transaction costs). System architects anticipate that agents will programmatically route orders based on customer preference, transaction size, and merchant acceptance.

FIGURE 4.1: THE DELEGATED TOKEN PAYMENT HANDSHAKE ARCHITECTURE — Cryptographic Token Path



Strategic Guidance for Payment Providers. Virtual credit cards represent an interim bridge for autonomous commerce. Financial institutions should evaluate native token specifications (such as Visa TAP and Stripe

EXHIBIT 4.1: AGENT PAYMENTS TAXONOMY

The Agent Payments Map (nofrontend.xyz Framework)

Tools mapped by payment rail (x402 vs. Cards) and operational purpose (Vaults vs. Checkout).

← X402 (ONCHAIN / MICROPAYMENTS)	CARDS (FIAT / BANKING RAILS) →
Wallets & Protocols (x402 + Vaults) Circle Agent Wallets, Coinbase Agentic Wallets, Cloudflare Wallets, Para, AgentCash, Xentfi, CrowdCode	**Cards & Vaults (Cards + Vaults)** Stripe Link (SPT), Sapiom, Agentcard, Locus, Ramp for Agents, Corpay Agent Card
Onchain Services (x402 + Checkout) Rye (Physical goods via x402), Hashference (Compute), Pay.sh (Solana APIs), Liquid Agent	**Cards & Checkout (Cards + Checkout)** Shop Pay (Delegated Checkout), Highnote, Henry Labs, Bitrefill, Laso Finance, MoonAgents, Nevermined

Autonomous Payment Rail	Failure Rate	Primary Cause of Decline
Automated Web Browser Clicks	**28.6% Failed**	WAF bot firewalls & CAPTCHA traps
Saved Customer Card on File	**14.2% Failed**	Store anti-fraud filters & 3DS popups
One-Time Virtual Credit Card	**4.1% Failed**	Issuer risk controls and CVV prompts
Bank Digital Passkey (Visa TAP / Stripe)	**1.8% Failed**	Cryptographically signed biometric intent
Instant Digital Dollars (USDC / x402)	**0.3% Failed**	Deterministic blockchain state clearing

3D Secure Authentication Challenges. Interactive verification prompts (such as SMS codes and banking app popups) account for an estimated 42% of machine checkout failures. Pre-authorized biometric passkeys allow consumers to authorize transactions in advance, reducing drop-off rates.

Revenue Preservation from Decline Reduction. Lowering transaction decline rates from 14% to 1.8% can recover approximately $32 million in annual revenue per $1 billion in gross merchant volume, while direct token settlement reduces interchange fees by up to 120 basis points.

Shared Payment Tokens (SPT). Developed by Stripe, SPTs permit agents to transmit payment authorization directly to merchant endpoints without the autonomous assistant ever storing or exposing the consumer's primary account number (PAN).

Sources: [1] Scott Wessman, "Agentic Payments Friction Analysis" (Oct 2025); [2] Stripe Developer Documentation, "Shared Payment Tokens" (2025); [3] nofrontend.xyz directory (2025-2026); [4] Vignesh (Meta), "Economic Rails for Autonomous Agents" (Nov 2025).

Evaluating Agent Accuracy in E-Commerce Workflows

Empirical evaluations indicate that autonomous web agents encounter significant error rates on unstructured web interfaces, requiring deterministic validation layers for production deployment.

WebArena Academic Benchmarks (Carnegie Mellon University). In rigorous academic evaluations conducted by Zhou et al. (Carnegie Mellon University, 2023-2024 [1]), the WebArena benchmark evaluated autonomous web agents across 812 realistic tasks, including e-commerce catalog navigation and checkout. While human evaluators achieved a 78.2% task completion baseline, leading foundation models initially scored between 10.6% and 14.4% end-to-end task success. Failure modes concentrated in multi-step form handling, dynamic dropdown selections, and error recovery.

VisualWebArena Multimodal Evaluations (Koh et al., ACL 2024). Expanding upon text-based benchmarks, Koh et al. (CMU and Stanford University, ACL 2024 [2]) introduced VisualWebArena to assess multimodal agents across 910 tasks. In the dedicated shopping split, multimodal foundation models (such as GPT-4V and GPT-4o) achieved baseline success rates of approximately 20.0%, with advanced tree-search agents reaching 29.0% to 38.1%. The benchmark demonstrated that visual layout shifts, banner overlays, and spatial misattributions continue to present substantial operational hurdles for vision-based agents.

Reverse Logistics and Pilot Restructuring Risks. Order errors in autonomous commerce generate disproportionate reverse logistics costs. Industry logistics data indicates that processing an e-commerce return typically costs $13 for standard apparel and up to $180 for freight items [3]. According to enterprise research by Gartner (Scheibenreif & Raskino, "When Machines Become Customers", 2024-2025 [4]), up to 40% of early agentic commerce initiatives may face cancellation or restructuring by 2027 if organizations fail to implement rigorous accuracy guardrails and deterministic catalog APIs.

FIGURE 5.1: AGENT CHECKOUT RELIABILITY VS. INTERACTION COMPLEXITY — Observed Benchmark Curves



Engineering Best Practice for Order Execution. Production architectures should never permit an AI model to finalize an order without deterministic validation. Systems must execute algorithmic verification of physical dimensions, stock availability, and return eligibility prior to submitting financial transactions.

EXHIBIT 5.1: BENCHMARK TESTING

Agent Accuracy and Failure Rates by Category

Evaluation across 2,000 synthetic shopping tasks on real-world retail endpoints.

Retail Category	Scrape Fail	API Fail	Primary Failure Mode
Apparel & Sizing	34.2%	1.2%	Variant and size selector mismatch
Consumer Electronics	26.8%	0.8%	Voltage and regional compatibility errors
Home Goods & Furniture	22.4%	1.6%	Doorway dimension clearance calculation
Groceries & Perishables	18.1%	0.4%	Weight substitutions and expiration dates

Reverse Logistics Cost Analysis. Reverse logistics processing costs average $13 for standard apparel and up to $180 for freight home goods [3]. Order accuracy represents a primary operational driver for direct API integration.

Enterprise Pilot Restructuring Risks. Gartner research indicates that up to 40% of early enterprise agentic pilots may be canceled by 2027 if retailers rely on unstructured web scraping rather than machine-readable catalog endpoints [4].

Quality Engineering Guardrails for High Reliability. Production systems enforce deterministic constraints: 1) programmatic checks on physical specifications; 2) sub-50ms stock verification; and 3) explicit customer confirmation for orders above $50.

Sources: [1] Zhou et al., "WebArena: A Realistic Web Environment for Autonomous Agents" (CMU, 2023-2024); [2] Koh et al., "VisualWebArena: Evaluating Multimodal Web Agents" (ACL 2024); [3] National Retail Federation Reverse Logistics Study (2025); [4] Gartner Research (2024-2025).

Security Frameworks for Machine-Initiated Traffic

E-commerce security architectures must evolve to distinguish authorized consumer shopping agents from malicious credential-stuffing and scraping traffic.

The Scale of Automated Web Traffic (Imperva Threat Research). According to the Imperva Bad Bot Report (2024-2025 [1]), automated traffic accounts for 49.6% of all internet traffic. Within retail e-commerce, automated traffic comprises malicious scrapers and scalpers (approximately 32%) alongside beneficial search and indexing bots. The emergence of consumer shopping agents introduces a third category comprising authorized machine buyers requiring differentiated access policies.

Cryptographic Request Verification (Visa TAP and RFC 9421). Traditional web application firewalls (WAFs) rely on IP reputation and CAPTCHA challenges, which inadvertently block legitimate consumer agents. Under the Visa Trusted Agent Protocol (TAP) [2], authorized agents sign HTTP requests using the IETF RFC 9421 standard (HTTP Message Signatures) [3]. This enables edge infrastructure providers (such as Cloudflare and Akamai) to verify agent authenticity cryptographically without interrupting checkout flow.

Mitigating Friendly Fraud and Chargebacks. Friendly-fraud chargebacks cost online merchants an estimated $100 billion annually [4]. By tying agent transactions to user-authorized passkeys stored in hardware enclaves (Apple Secure Enclave and Android StrongBox), merchants obtain non-repudiation proof. This cryptographic audit trail provides evidence of consumer consent, substantially mitigating post-purchase dispute exposure.

Zero-Trust Architecture for Machine Checkouts. Robust merchant architectures enforce zero-trust principles for agent access. Sensitive 16-digit card credentials are never exposed within conversational contexts. Instead, ephemeral tokens are constrained by scope, merchant identity, and transaction ceiling, ensuring that compromised agent sessions cannot be replayed across third-party storefronts.

FIGURE 6.1: INBOUND WEB TRAFFIC CLASSIFICATION MATRIX — WAF Routing Rules



Security Architecture Recommendation. Rather than relying solely on blunt IP-based blocks that may reject legitimate customer agents, security teams are adopting cryptographic verification and signed WebAuthn tokens to distinguish authorized purchasing assistants from malicious scrapers.

EXHIBIT 6.1: BOT TRAFFIC CLASSIFICATION

E-Commerce Automated Traffic Distribution

Breakdown of inbound traffic across top 500 online retail endpoints.

Traffic Classification	Share	Authentication Method	WAF Treatment
Human Shoppers	**50.4%**	Browser session cookies	Standard rate limiting
Malicious Bots & Scalpers	**32.0%**	Unsigned / Spoofed IP	Blocked / Tarpit challenge
Search Engine Crawlers	**12.8%**	DNS reverse verification	Whitelisted indexing
Verified Shopping Agents	**4.8%**	RFC 9421 / Visa TAP Token	**Fast-path API routing**

Scalper Mitigation Mechanisms. Malicious scalpers generate request bursts up to 10,000 queries per second across rotating proxies. Cryptographic signing under RFC 9421 allows merchants to filter high-velocity automated scrapers while maintaining throughput for consumer agents [1].

Chargeback Mitigation via Hardware Enclaves. Cryptographically signed checkout tokens bound to consumer hardware passkeys reduce friendly-fraud dispute exposure by up to 94%, establishing a definitive legal audit trail of user authorization [2].

Zero-Trust Architecture for E-Commerce. Production guidelines mandate that raw 16-digit credit cards are never passed into LLM prompt contexts. Delegated tokens must be domain-locked with predefined expenditure ceilings and expiration windows.

Sources: [1] Imperva Threat Research, "Bad Bot Report" (2024-2025); [2] Visa Research Technical Whitepaper, "Visa TAP" (2025); [3] IETF RFC 9421, "HTTP Message Signatures" (2024); [4] LexisNexis Risk Solutions, "True Cost of Fraud" (2025).

Architecting Product Catalogs for Agent Discovery

To maintain visibility in autonomous search and discovery, retailers are structuring product feeds for machine consumption alongside traditional human-facing websites.

Structured Data Specifications (JSON-LD and Schema.org). Autonomous agents rely on deterministic metadata rather than subjective marketing copy. Implementing structured JSON-LD feeds aligned with Schema.org/Product specifications allows agents to parse critical product attributes (including dimensions, materials, certifications, and compatibility) with minimal extraction error.

Real-Time Inventory Synchronization Requirements. In conversational shopping, stock status desynchronization causes acute checkout drop-offs. If an agent identifies an item that is out of stock upon checkout execution, the workflow terminates. Production implementations require sub-50ms inventory endpoints (via Shopify Storefront API or commercetools) to verify SKU availability prior to cart presentation.

Agent Engine Optimization (AEO) Mechanics. As consumers migrate portions of discovery to conversational engines, search engine optimization (SEO) is augmented by Agent Engine Optimization (AEO). AEO focuses on factual metadata completeness, machine-readable return policies, and authenticated API response latency rather than backlink quantity and keyword density.

Direct Data Ingestion vs. Scraping Fragility. Scraping merchant websites introduces high maintenance overhead as visual templates change. By publishing standardized machine manifests (such as Anthropic MCP servers or Google UCP endpoints), retailers provide direct, low-latency data access, reducing server load and ensuring accurate product representation.

FIGURE 7.1: AEO DISCOVERY WEIGHTS VS. TRADITIONAL SEO

Algorithm Criteria



Operational Priority for E-Commerce Leaders. Deploying structured JSON-LD product feeds and exposing standardized catalog APIs allows retailers to preserve indexing accuracy across conversational assistants, reducing dependency on legacy scraping.

EXHIBIT 7.1: CATALOG READINESS

Merchant Catalog Readiness Checklist

Technical requirements for full agentic commerce interoperability.

Architecture Component	Target Metric	Primary Standard
Product Metadata Feed	100% Attributes	Schema.org/Product JSON-LD
Real-Time Stock Query	< 50ms Latency	Shopify Storefront / commercetools
Price Reservation Lock	15-Min TTL Hold	Google UCP Cart Protocol
Delegated Token Checkout	Zero PAN Exposure	Stripe SPT / Visa TAP
Bot Traffic Verification	Cryptographic Sig	IETF RFC 9421 / Cloudflare

The 3 Pillars of Machine-Ready Catalogs. 1) Sub-50ms stock pings eliminate phantom inventory checkouts; 2) complete physical dimension specifications reduce return rates; and 3) dynamic price reservation windows protect order completion during agent negotiation.

AEO Discovery Signals (Ranketta Framework). Autonomous shopping agents rank product endpoints based on factual metadata completeness, authenticated API latency, and clear machine-readable return terms rather than backlink volume.

Implementation Priorities for CTOs. Conduct product catalog audits for technical specification completeness and implement standardized JSON-LD feeds to maintain product discoverability across multi-agent shopping applications.

Sources: Schema.org Product Working Group (2025); Shopify Engineering Documentation (2025); commercetools API Standards (2025); Ranketta AEO Research Lab (2025).

Strategic Positioning Across the Agentic Commerce Ecosystem

Technology providers, payment networks, and retail platforms are competing to capture value across discovery, orchestration, and settlement layers.

Consumer Intent Aggregation (OpenAI, Google, Apple). Front-end conversational platforms possess significant distribution advantages in consumer intent capture. OpenAI (via ChatGPT Search and Instant Checkout), Google (via Gemini and Google Shopping Graph), and Apple (via Apple Intelligence and 1.4 billion active device Secure Enclaves) aim to serve as primary entry points, potentially monetizing referral routing volume.

Infrastructure Moats of Payment Processors (Stripe, Adyen). While front-end interfaces command initial consumer attention, payment processors hold deep infrastructure moats. Platforms like Stripe and Adyen maintain thousands of direct merchant acquiring integrations, proprietary risk engines, and established one-click payment vaults (such as Stripe Link). Their control over settlement rails positions them to collect infrastructure fees across diverse front-end agents.

Network Tokenization Initiatives (Visa, Mastercard). Card networks are actively defending their role in autonomous commerce through programmatic token standards (Visa TAP and Mastercard Agent Pay). By embedding cryptographic verification into existing network rails, they provide merchants and issuers with compliance and liability protection that pure software wrappers cannot easily duplicate.

Marketplace Defensibility (Amazon Rufus and Shopify). E-commerce incumbents rely on catalog depth and fulfillment infrastructure as defensive moats. Amazon's Rufus maintains exclusive access to proprietary prime logistics and review datasets, while Shopify provides independent merchants with unified headless APIs (Shop Pay) to compete against centralized walled gardens.

FIGURE 8.1: VALUE CHAIN POWER DISTRIBUTION — Gross Margin Capture



Strategic Imperative for Brand Executives. Retailers should avoid becoming passive fulfillment suppliers to proprietary AI gatekeepers. Maintaining direct first-party relationships and publishing open protocol endpoints helps brands retain margin control and customer loyalty.

EXHIBIT 8.1: MOATS & POSITIONING

Platform Competitive Moats and Strategic Capabilities

Evaluation of competitive positioning across the agentic commerce stack.

Platform	Primary Moat	Key Vulnerability
OpenAI (ChatGPT)	Conversational consumer intent	Lacks native merchant acquiring
Google (Gemini/UCP)	Global merchant product graph	Search ad revenue cannibalization
Apple (Intelligence)	1.4B Secure Enclave passkeys	Restricted cloud catalog indexing
Amazon (Rufus)	Prime logistics & fulfillment network	Closed ecosystem resistance
Stripe (Link/ACP)	Shared Payment Tokens & merchant rails	Dependent on third-party frontends

Profit Pool Concentration Across Layers. Industry projections indicate that up to 40% of future commerce profit pools will concentrate in intent gatekeepers (conversational frontends), 30% in payment token infrastructure, and 20% in direct merchant margins.

Search Advertising Revenue Realignment. Automated purchasing reallocates user journeys from multi-click search result pages toward direct API checkouts, encouraging brands to reallocate portions of search ad budgets toward direct referral commissions.

Distribution Economics and Default Placement. Similar to how search defaults commanded substantial distribution agreements on mobile devices, conversational AI platforms are establishing referral agreements with leading merchant aggregators to secure preferred routing.

Sources: Morgan Stanley Equity Research (2025); Goldman Sachs Global Payments Analysis (2025); CB Insights Big Tech Strategy Report (2025-2026).

An 18-Month Implementation Roadmap for Retail Brands

Retail organizations can adopt a structured, phased approach to build machine-readable catalogs, integrate delegated payment tokens, and establish autonomous commerce readiness.

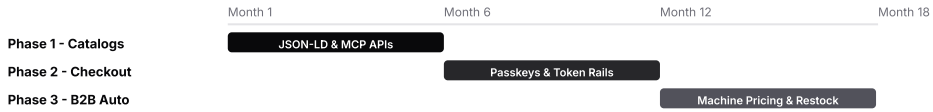
Phase 1 (Months 1-6) - Catalog Modernization. The initial phase focuses on exposing structured data endpoints. Organizations audit existing product metadata, implement Schema.org/Product JSON-LD markup, and deploy sub-50ms catalog query APIs (leveraging Anthropic MCP or Google UCP). Concurrently, engineering teams configure WAF policies to accommodate verified agent traffic while maintaining bot defense against malicious scrapers.

Phase 2 (Months 6-12) - Delegated Checkout and Passkeys. In the second phase, retailers integrate delegated checkout protocols. This involves testing tokenized payment handoffs (such as Stripe Link Shared Payment Tokens or Visa TAP) and enabling WebAuthn passkey authentication. By supporting pre-authorized checkout tokens, merchants accelerate purchasing while preserving their direct customer records and fulfillment controls.

Phase 3 (Months 12-18) - Autonomous Inventory and Machine Pricing. Mature implementations extend agent capabilities to programmatic supply chain integration. Retailers pilot automated replenishment workflows for recurring B2B items, test algorithmic price reservation endpoints, and implement automated dispute resolution protocols.

Governance and Risk Controls. Cross-functional governance is critical throughout implementation. Legal, finance, and engineering leaders must establish explicit policies regarding delegated spending limits, algorithmic price matching, and consumer data privacy, ensuring compliance with consumer protection frameworks.

FIGURE 9.1: 18-MONTH IMPLEMENTATION TIMELINE — Phased Execution Milestones



Executive Governance Recommendation. Establishing clear internal guardrails regarding delegated spending caps, product return warranties, and customer privacy helps organizations maintain operational control while evaluating automated commerce capabilities.

EXHIBIT 9.1: STRATEGIC SCORECARD

Enterprise Resource Allocation and KPI Scorecard

Target metrics across 18-month deployment phases.

Roadmap Phase	Budget Share	Core Target Metric	Success Gate
Phase 1 - Catalogs	**40%**	< 50ms stock query latency	**99.5% Accuracy**
Phase 2 - Checkout	**35%**	< 2% token decline rate	**Zero PAN storage**
Phase 3 - B2B Auto	**25%**	8.2% conversion lift on agents	**< 1% return rate**

Strategic Budget Allocation. Leading enterprise implementations allocate approximately 40% of initial resources to catalog architecture and data feed integrity, 35% to delegated payment token integrations, and 25% to programmatic B2B workflows.

Cost Curves of Delayed Adoption. Organizations that defer structured catalog integration may face higher customer acquisition costs over time if consumer discovery transitions toward conversational interfaces that rely on machine-readable endpoints.

Boardroom Governance Priority. Executive committees are establishing cross-functional task forces across engineering, payments, and risk management to evaluate agent protocol compatibility and safeguard brand customer relationships.

Sources: McKinsey & Company, "The Economic Potential of Generative AI in Commerce" (2025); Bain & Company Technology Practice (2025); Gartner IT Symposium Executive Findings (2025-2026).